SUMMER BAHAM / HANDS-ON IT SUPPORT LAB

CISCO ROUTER CONFIGURATION & INTER-NETWORK ROUTING

Packet Tracer | Cisco 1941 | IPv4 | Directly connected routes

VERIFIED LAB NOVEMBER 2026

PROJECT OVERVIEW

Verified an IPv4 routing lab connecting two endpoint networks through a Cisco 1941 router. Documented interface addresses, static endpoint settings, connected routes, and successful ping tests in both directions.

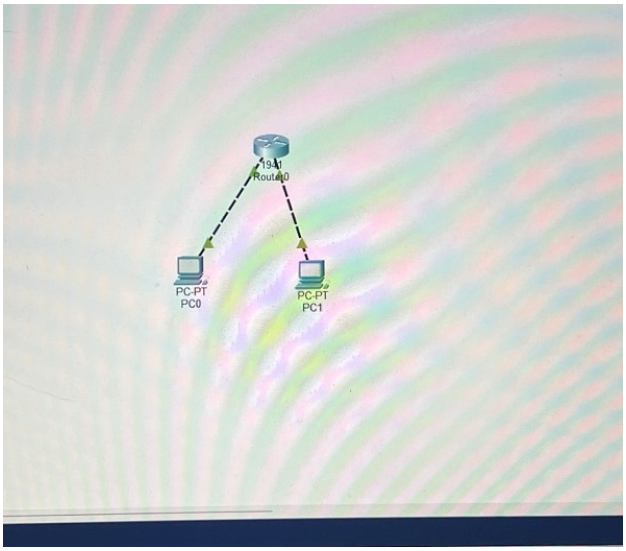


Figure 1 | Cisco Packet Tracer topology: Router0 connected directly to PC0 and PC1.

RESULT **Bidirectional IPv4 connectivity verified with 0% loss on final tests.**

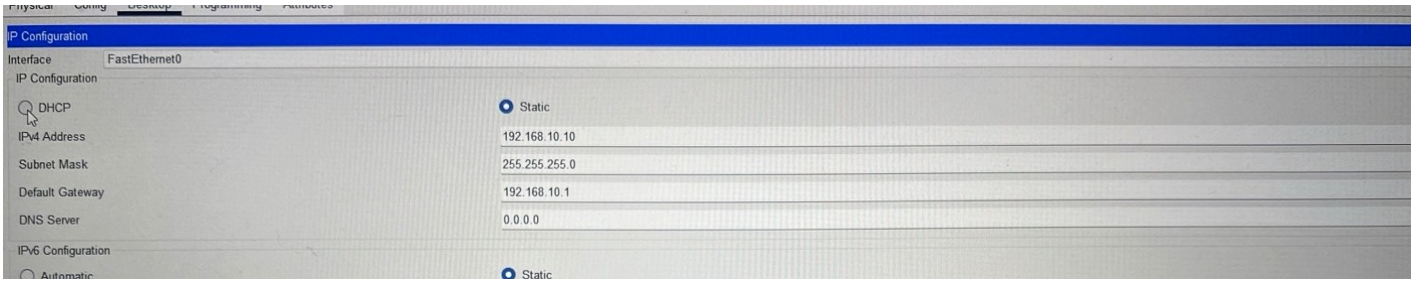
LAB SETUP

Two IPv4 Networks

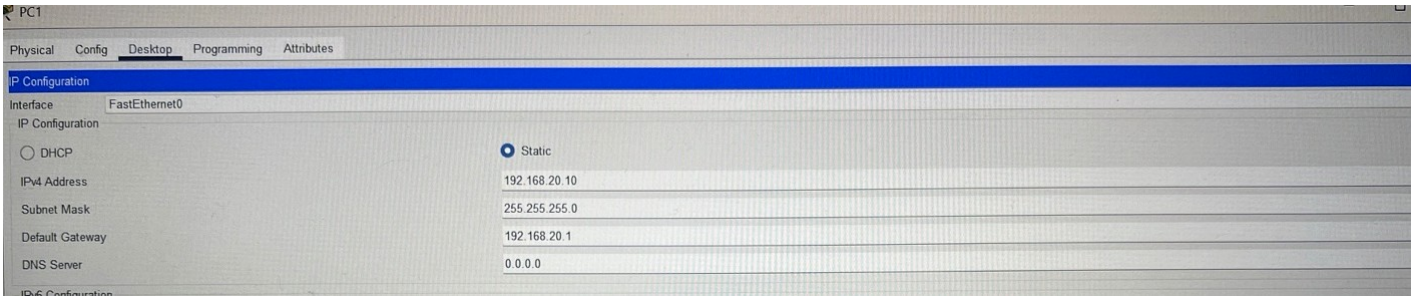
Router0 uses two Ethernet interfaces to route traffic between separately addressed /24 networks.

Component	IPv4 address	Subnet / gateway
Router0 G0/0	192.168.10.1	192.168.10.0/24
PC0	192.168.10.10	255.255.255.0 / 192.168.10.1
Router0 G0/1	192.168.20.1	192.168.20.0/24
PC1	192.168.20.10	255.255.255.0 / 192.168.20.1

EVIDENCE 01 / PC0 IPv4 CONFIGURATION



EVIDENCE 02 / PC1 IPv4 CONFIGURATION



Both screenshots show static IPv4 addressing. DNS is 0.0.0.0; name resolution was not part of this routing verification.

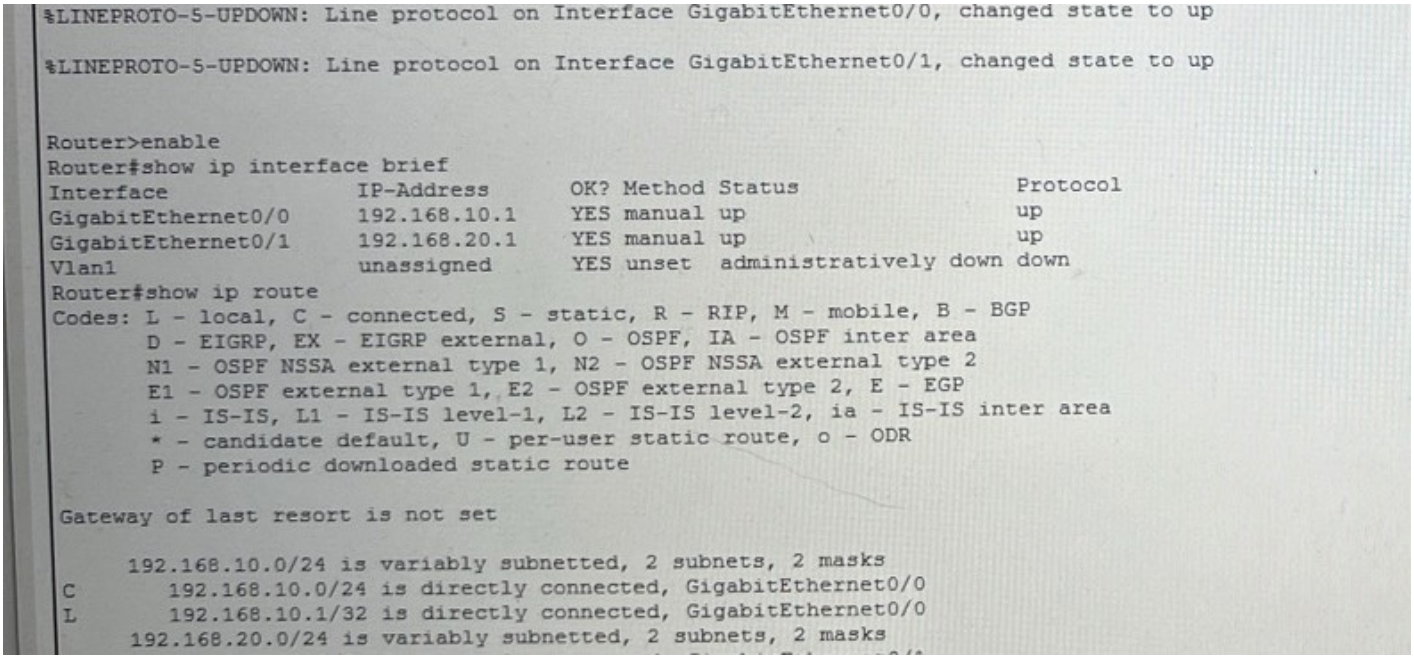
CISCO IOS

Interface State and Routing Table

Router0 was inspected using show ip interface brief and show ip route.

INTERFACE STATE

Interface	IPv4 address	Status / protocol
GigabitEthernet0/0	192.168.10.1	up / up
GigabitEthernet0/1	192.168.20.1	up / up
Vlan1	unassigned	administratively down / down



```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up


Router>enable
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     192.168.20.1    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
```

The routing table lists C routes to 192.168.10.0/24 via G0/0 and 192.168.20.0/24 via G0/1; L routes identify the router interface /32 addresses.

No gateway of last resort was set. A default route was not required for the two networks connected directly to this router.

TEST 01

PC1 to PC0: Forward Connectivity

Source 192.168.20.10 -> Destination 192.168.10.10

INITIAL TEST

Sent 4 / Received 3 / Lost 1 (25% packet loss)

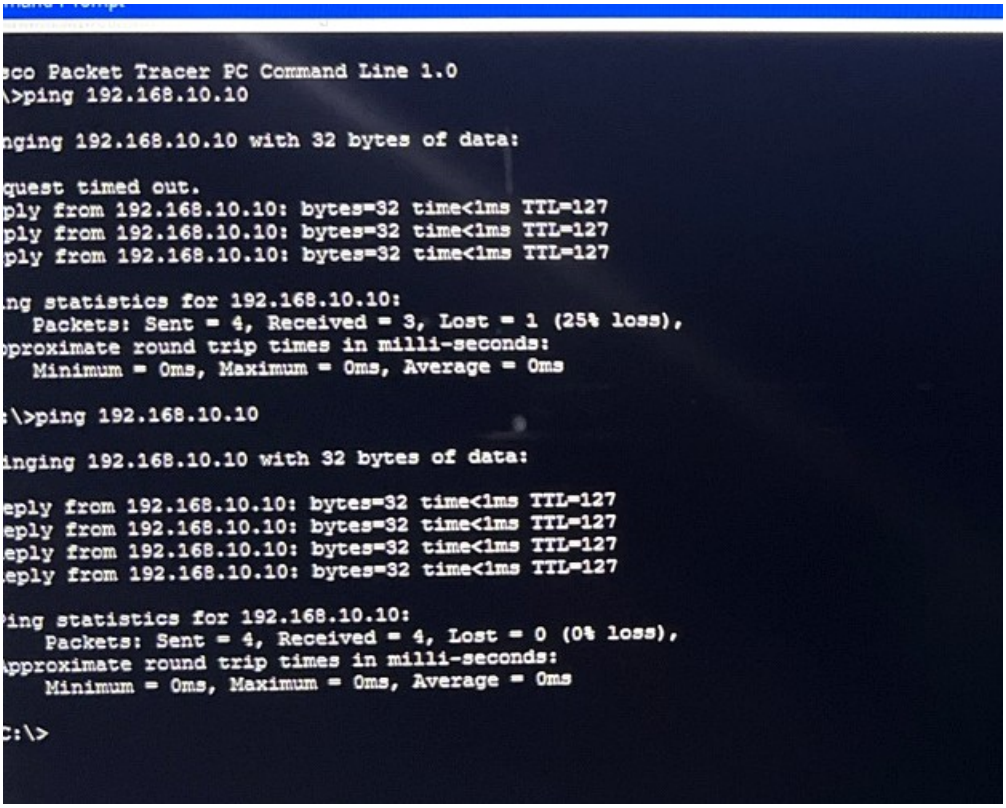
The first request timed out; three replies were successful.

REPEAT TEST

Sent 4 / Received 4 / Lost 0 (0% packet loss)

The repeat ping returned four replies, confirming reachability.

EVIDENCE 04 / PC1 COMMAND PROMPT



Interpretation: the first timeout may have been caused by initial ARP resolution. The screenshot alone does not establish the exact cause.

TEST 02

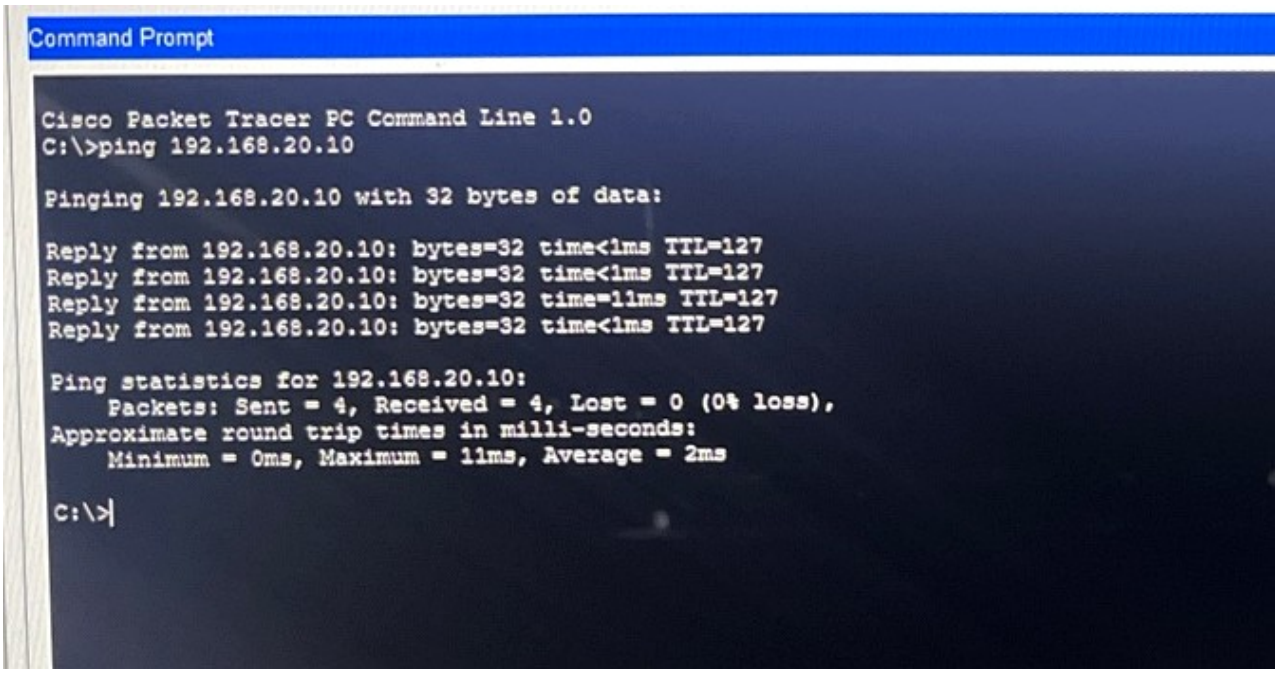
PC0 to PC1: Reverse Connectivity

Source 192.168.10.10 -> Destination 192.168.20.10

4 of 4 replies received

0% packet loss | TTL 127 | Reverse direction successful

EVIDENCE 05 / PC0 COMMAND PROMPT



FINAL RESULT

Both endpoints reached each other across the two routed IPv4 networks. The router interface state and connected routes were verified in Cisco IOS. The tests demonstrate functioning local inter-network routing in the Packet Tracer simulation.

SCOPE AND REPRODUCIBILITY

Evidence is based on the screenshots captured during the lab. No external internet routing, dynamic routing protocol, firewall policy, or physical hardware deployment was tested. The editable .pkt file should be saved separately from this PDF.

Documentation prepared for the Summer Baham IT Support Engineering & Cybersecurity portfolio.